

Shifted-exponent transport for the phase expansion

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Abstract

Unit 200 (programme A, step 2). The j -th Taylor term of the phase factor $e^{\beta N u^k \xi(u)}$ is $(\beta N)^j / j! \cdot \xi(u)^j u^{jk}$, which turns the dressed normal integral with exponents h into one with exponents $h + jk$ and amplitude $\xi^j \eta$. We record that the shift $(h, \lambda) \mapsto (h + jk, \lambda + j/2)$ leaves the entire face structure invariant — the minimiser set J , the multiplicity m , the face projection and the residual weight $\prod_{i \notin J} u_i^{h_i - 2k_i \lambda}$ — and only replaces $\Gamma(\lambda) \beta^{-\lambda}$ by $\Gamma(\lambda + j/2) \beta^{-\lambda - j/2}$ in the face constant. The j -th term is then packaged as a limit in the chart variable N (monomial parameter N^2):

$$\frac{N^j \int_{(0,1]^d} \eta(u) u^{h+jk} e^{-\beta N^2 u^{2k}} du}{N^{-2\lambda} (\log N)^{m-1}} \longrightarrow 2^{m-1} \text{amplitudeCoeff}(h + jk, k, \lambda + j/2, \beta)(\eta),$$

by the continuous-amplitude theorem of unit 186 and the square-parameter transport of unit 193.
Zero sorry/axiom; 9077 jobs.

1 The things formalised

```
def phaseShift (h k : Fin d → ) (j : ) : Fin d → := fun i => h i + j * k i
theorem prod_pow_phaseShift : i, x i ^ phaseShift h k j i = ( i, x i ^ h i) * ( i,
  x i ^ k i) ^ j
theorem ratioExp_phaseShift (hk) : ratioExp (phaseShift h k j) k i = ratioExp h k i + j
  / 2
theorem multCount_phaseShift : multCount (ratioExp (phaseShift h k j) k) (1 + j/2) =
  multCount (ratioExp h k) 1
theorem faceProj_phaseShift : faceProj (phaseShift h k j) k (1 + j/2) = faceProj h k 1
theorem residualWeight_phaseShift : residualWeight (phaseShift h k j) k (1 + j/2) =
  residualWeight h k 1
theorem faceLeadConst_phaseShift : faceLeadConst (phaseShift h k j) k (1 + j/2)
  = Γ (1 + j/2) * ^(-(1 + j/2)) / (multCount (ratioExp h k) 1 - 1)! * i, if
  ratioExp h k i = 1 then 1/(2 k i) else 1
theorem amplitudeCoeff_phaseShift : amplitudeCoeff (phaseShift h k j) k (1 + j/2)
  = Γ (1 + j/2) * ^(-(1 + j/2)) / (m-1)! * ( ...) * u in unitBox d, (faceProj h k
  1 u) * residualWeight h k 1 u
theorem shifted_term_tendsto (hmin) (hatt) (h : Continuous ) (j : ) :
  Tendsto (fun N => N^j * ( x in unitBox (d+1), x * (( i, x i ^ phaseShift h k j
  i)
    * exp (-( * N^2 * i, x i ^ (2 k i)))))) / (N^(-(2l)) * log N ^ (multCount
  (ratioExp h k) 1 - 1)))
  atTop ( (2 ^ (multCount (ratioExp h k) 1 - 1) * amplitudeCoeff (phaseShift h k
  j) k (1 + j/2) ))
```

2 Mathematics

$\text{ratioExp}(h + jk, k, i) = \frac{h_i + jk_i + 1}{2k_i} = \frac{h_i + 1}{2k_i} + \frac{j}{2}$, so $\text{ratioExp}(h + jk, k, i) = \lambda + j/2 \iff \text{ratioExp}(h, k, i) = \lambda$: the minimisers, hence J, m , the face projection and the indicator structure of every face object, are unchanged. The residual exponents $h_i + jk_i - 2k_i(\lambda + j/2) = h_i - 2k_i\lambda$ are literally unchanged. The bookkeeping $N^j (N^2)^{-(\lambda + j/2)} = N^{-2\lambda}$ and $(\log N^2)^{m-1} = 2^{m-1}(\log N)^{m-1}$ gives the 2^{m-1} ; the $\beta^j/j!$ of the Taylor coefficient is left to the assembly unit.

3 Lean notes

- `shiftExp` was already taken (`MixedBlockEnvelope.lean`, a one-coordinate shift by a unit vector); the phase shift is `phaseShift`. The "already declared" error appears only at the second definition, but every later use silently resolved to the OLD constant and produced "application type mismatch" cascades — check the first error only.
- Rewriting under the condition of an `if` with `rw` fails (the `Decidable` instance depends on the proposition); `simp only [iff_lemma]` handles the instance transport, or use `by_cases` with `if_pos/if_neg`.
- `Real.rpow_neg hN.le` without an explicit exponent rewrites only the first instantiation it meets; a second occurrence with a different exponent needs its own `Real.rpow_neg hN.le (j : )` before `Real.rpow_natCast` can fire.
- `filter_upwards` already beta-reduces the pointwise goal of a `Tendsto.congr'`; a following `simp only []` errors with "no progress".

4 Next

Unit 201: the phase-dressed normal moment $S_{p/2}^{(\beta)}(a) = \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta s a} ds$: integrability, Gaussian moments $\int_0^\infty s^{p+j-1} e^{-\beta s^2} ds = \frac{1}{2} \Gamma(\lambda + j/2) \beta^{-\lambda-j/2}$, and the finite Taylor approximation with the uniform tail of unit 199.